

Version 2.1 Reproducibility and Consistency Audit

Project: *When Does Option-Implied Variance Add to Physical Forecasts? Evidence from S&P 500 Leveraged ETF Drag*

Status: PASS

Automated checks passed: 1,080 of 1,080

1. What this audit establishes

The audit checks whether the manuscript, supplement, tables, figures, code, and frozen inputs are internally consistent and reproducible. It verifies forecast timing, recalculates the principal statistics from row-level forecasts, reruns the analysis in a clean directory, checks anonymity and wording, and validates the V2.1 hardening outputs.

The audit does **not** establish mathematical certainty, causal identification, external validity beyond the implemented S&P 500/VIX sample, or legal permission to redistribute third-party data.

2. Frozen inputs

- Market-data SHA-256: 54e965d435da3b26ecc50cb6df7bc193d6b66a9b202837b9d2002a9e7384c715
- Rate-data SHA-256: b73beaedaaf92394adf94ed7de7660ccf52e8ca5ae63a8cb6c69edb0ad29f461
- Complete sample: 4,023 trading dates, January 4, 2010 through December 30, 2025
- Required market series: SPY, SSO, UPRO, SDS, SPXU, VIX, and VIX3M
- Additional series: three-month Treasury-bill rate

Automated checks for date uniqueness, missing values, positivity, sample range, selected shock dates, and approximate fund betas passed.

3. Recomputed core results

Horizon	Raw option	MZ option	HAR(4)	HAR + option + term
5	0.352	0.301	0.114	0.254
22	0.184	0.179	0.142	0.192
66	-0.204	0.063	0.025	0.059

Every baseline out-of-sample R2 value was recomputed from the row-level target and forecast columns and matched the result files within numerical tolerance below 5e-13.

At 66 days, the independently recomputed MZ-versus-raw squared-loss comparison reproduced HAC $t = 2.3320$ and $p = 0.0197$. The future-data perturbation test changed all market observations after December 31, 2019 and produced zero change in every forecast dated on or before that cutoff.

4. V2.1 hardening checks

- Under mean absolute error, MZ improves on raw option variance at 66 days (HAC $p = 0.000004$).
- Under QLIKE with the common 1-percent-of-mean-target floor, MZ does not improve on raw option variance (mean gain = -0.008487; HAC $p = 0.925$). The manuscript therefore makes no loss-uniform dominance claim.

- The 66-day squared-loss HAC p-values range from 0.019 to 0.028 across the reported bandwidths; circular-block p-values range from 0.020 to 0.025.
- The raw 66-day option forecast averages 1.431 times mean realized variance; MZ averages 0.883. The squared-bias share of MSE falls from 0.061 to 0.006, while correlation does not increase. This supports a level-calibration interpretation.
- Multiplying idealized variance targets and forecasts by leverage loadings 1, 3, or 6 changes out-of-sample R2 by at most 4.441e-16. Thus, leverage scaling alone cannot explain fund-level R2 differences.
- Actual-calendar-day scaling leaves the 66-day raw and MZ results nearly unchanged.

5. Clean-room reproduction and artifact checks

The core and hardening analyses were run in a new output directory using only packaged code and frozen inputs. All deterministic CSV outputs matched in shape, columns, text fields, and numerical values within 5e-12. The four PNG figures were byte-identical.

The audit also checks:

- anonymous source and PDF contain no author names or emails;
- no obsolete causal, novelty, HAR-RV, VIX9D-implementation, placeholder, TODO, or surname-error language remains;
- the manuscript explicitly reports the QLIKE qualification, SPY-SPX proxy basis, ex post status of regime splits, and missing intraday/NAV extensions;
- LaTeX table fragments have valid row terminators;
- all final DOCX and PDF artifacts exist and contain the revised V2.1 content.

6. Scientific limitations that remain

The following gaps require new verified or licensed data and are not described as completed:

1. Intraday returns for high-frequency realized variance.
2. Historical sponsor NAV series for a cleaner compounding target.
3. Removal of the SPY-versus-SPX option-index proxy basis.
4. A maturity-closer VIX9D implementation at the 5-day horizon.
5. Nasdaq-100/VXN or other-market replication.
6. Direct investor holding-period evidence for the 66-day risk-monitoring horizon.

The manuscript handles these gaps by narrowing its population, labeling the fund outcome market-price drag, treating 66 days as a stress and risk-monitoring horizon, and avoiding claims beyond the implemented evidence.

7. Conclusion

Version 2.1 is internally consistent and reproducible from the frozen inputs. No unresolved numerical mismatch, stale table, look-ahead failure, anonymity leak, unsupported completed-test claim, placeholder, or broken final artifact was found. This conclusion is limited to the supplied data and implemented analyses.